APPLICATIONS OF DIFFERENTIATION

1. Maximum and Minimum Values

Q1. Find the critical numbers of the function $\mathbf{f(x) = x^3 + 6x^2 - 7x}$.

S1

1. **Find** $f'(x)$: $f'(x) = 3x^2 + 12x - 7$. 2. **Set** $f'(x) = 0$: Using the quadratic formula: $x = \dfrac{-12 \pm \sqrt{144+84}}{6}$. The critical numbers are $\mathbf{x = \dfrac{-12 \pm \sqrt{228}}{6}}$.

Q2. Find the critical numbers of the function $\mathbf{f(x) = x^4 - 2x^2 + 3}$ on the interval $[\mathbf{-2, 3}]$.

S2

1. **Find** $f'(x)$: $f'(x) = 4x^3 - 4x = 4x(x^2-1) = 4x(x-1)(x+1)$. 2. **Set** $f'(x) = 0$: $\mathbf{x = 0, x = 1}$, and $\mathbf{x = -1}$. All are in $[-2, 3]$.

Q3. Find the absolute maximum and minimum values of $\mathbf{f(x) = x^3 - 3x + 1}$ on the interval $[\mathbf{0, 3}]$.

S3

1. **Critical points**: $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$. Thus the critical numbers are $x = -1$ and $x = 1$, but only $x = 1$ lies in the interval $[0, 3]$. 2. **Evaluate**: $f(0) = 1$, $f(1) = -1$, $f(3) = 19$.

The absolute maximum is **19** (at $x = 3$), and the absolute minimum is $-\mathbf{1}$ (at $x = 1$).

2. The Mean Value Theorem

Q4. State the main conclusion of the Mean Value Theorem (MVT).

S4

If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists a number $\mathbf{c} \in (\mathbf{a}, \mathbf{b})$ such that

$$\mathbf{f'(c) = \frac{f(b) - f(a)}{b - a}}.$$

Q5. Prove that the equation $\mathbf{x^3 + 15x - c = 0}$ has at most one real root.

S5

Let $f(x) = x^3 + 15x - c$. Then $f'(x) = 3x^2 + 15$. Since $f'(x) > 0$ for all x, f is strictly increasing, so it can have at most one real root.

3. Indeterminate Forms and L'Hospital's Rule

Q6. Evaluate the limit: $\displaystyle\lim_{\mathbf{x \to 0}} \mathbf{\frac{e^{4x} - 1 - 4x}{x^2}}$.

S6

1. **Indeterminate Form (IF):** Substituting $x = 0$ gives $\dfrac{1-1-0}{0} = \dfrac{0}{0}$.

2. **Apply L'Hôpital's Rule (1st time):**

$$\lim_{x \to 0} \frac{e^{4x} - 1 - 4x}{x^2} = \lim_{x \to 0} \frac{4e^{4x} - 4}{2x} = \lim_{x \to 0} \frac{4(e^{4x} - 1)}{2x}.$$

Yine $\frac{0}{0}$.

3. **Apply L'Hôpital's Rule (2nd time):**

$$\lim_{x \to 0} \frac{4(e^{4x} - 1)}{2x} = \lim_{x \to 0} \frac{16e^{4x}}{2} = 8.$$

Thus,

$$\boxed{\lim_{x \to 0} \frac{e^{4x} - 1 - 4x}{x^2} = 8}.$$

Q7. Evaluate the limit: $\lim_{x \to 0} \dfrac{e^x - 1 - x}{x^2}$.

S7

- **IF:** Substituting $x = 0$ gives $\dfrac{1 - 1 - 0}{0} = \dfrac{0}{0}$.
- **1st L'Hôpital:**

  $$\lim_{x \to 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \to 0} \frac{e^x - 1}{2x},$$

  which is still $\dfrac{0}{0}$.
- **2nd L'Hôpital:**

  $$\lim_{x \to 0} \frac{e^x - 1}{2x} = \lim_{x \to 0} \frac{e^x}{2} = \frac{1}{2}.$$

Therefore,

$$\boxed{\lim_{x \to 0} \frac{e^x - 1 - x}{x^2} = \frac{1}{2}}.$$

Q8. Evaluate the limit: $\lim_{x \to 0} \dfrac{x - \sin x}{x^3}$.

S8

- **IF:** Substituting $x = 0$ gives $\dfrac{0 - 0}{0} = \dfrac{0}{0}$.
- **1st L'Hôpital:**

  $$\lim_{x \to 0} \frac{x - \sin x}{x^3} = \lim_{x \to 0} \frac{1 - \cos x}{3x^2},$$

  still $\dfrac{0}{0}$.
- **2nd L'Hôpital:**

  $$\lim_{x \to 0} \frac{1 - \cos x}{3x^2} = \lim_{x \to 0} \frac{\sin x}{6x},$$

  still $\dfrac{0}{0}$.
- **3rd L'Hôpital:**

  $$\lim_{x \to 0} \frac{\sin x}{6x} = \lim_{x \to 0} \frac{\cos x}{6} = \frac{1}{6}.$$

Thus,

$$\boxed{\lim_{x \to 0} \frac{x - \sin x}{x^3} = \frac{1}{6}}.$$

Q9. Evaluate the limit: $\lim_{x \to 0} \dfrac{\cos x - 1 + x^2/2}{x^4}$.

S9

- **IF:** Substituting $x = 0$ gives $\dfrac{1-1+0}{0} = \dfrac{0}{0}$.
- Let $N(x) = \cos x - 1 + \dfrac{x^2}{2}$ and $D(x) = x^4$.
- **1st derivatives:**
$$N'(x) = -\sin x + x, \qquad D'(x) = 4x^3.$$
At $x \to 0$: $N'(0) = 0$, $D'(0) = 0 \Rightarrow \dfrac{0}{0}$.
- **2nd derivatives:**
$$N''(x) = -\cos x + 1, \qquad D''(x) = 12x^2.$$
Again $N''(0) = 0$, $D''(0) = 0$.
- **3rd derivatives:**
$$N'''(x) = \sin x, \qquad D'''(x) = 24x,$$
still $0/0$ at $x = 0$.
- **4th derivatives:**
$$N^{(4)}(x) = \cos x, \qquad D^{(4)}(x) = 24.$$
Thus,
$$\lim_{x\to 0} \frac{\cos x - 1 + x^2/2}{x^4} = \lim_{x\to 0} \frac{N^{(4)}(x)}{D^{(4)}(x)} = \frac{\cos 0}{24} = \frac{1}{24}.$$

So
$$\boxed{\lim_{x\to 0} \frac{\cos x - 1 + x^2/2}{x^4} = \frac{1}{24}}.$$

Q10. Evaluate the limit: $\displaystyle\lim_{t\to 0} \frac{\cosh t - 1}{t^2}$.

S10

- **IF:** Substituting $t = 0$ gives $\dfrac{\cosh 0 - 1}{0} = \dfrac{1-1}{0} = \dfrac{0}{0}$.
- **1st L'Hôpital:**
$$\lim_{t\to 0} \frac{\cosh t - 1}{t^2} = \lim_{t\to 0} \frac{\sinh t}{2t},$$
still $\dfrac{0}{0}$.
- **2nd L'Hôpital:**
$$\lim_{t\to 0} \frac{\sinh t}{2t} = \lim_{t\to 0} \frac{\cosh t}{2} = \frac{1}{2}.$$

Therefore,
$$\boxed{\lim_{t\to 0} \frac{\cosh t - 1}{t^2} = \frac{1}{2}}.$$

Q11. Evaluate the limit: $\displaystyle\lim_{x\to 0^+} x \ln x$.

S11

1. **IF:** As $x \to 0^+$, $x \to 0$ and $\ln x \to -\infty$, so the product is of the indeterminate form $0 \cdot (-\infty)$.
2. **Rewrite as a quotient:**
$$x \ln x = \frac{\ln x}{1/x}.$$
Now as $x \to 0^+$, $\ln x \to -\infty$ and $1/x \to \infty$, so we have $\dfrac{-\infty}{\infty}$.
3. **Apply L'Hôpital's Rule:**
$$\lim_{x \to 0^+} \frac{\ln x}{1/x} = \lim_{x \to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \to 0^+} (-x) = 0.$$

Hence
$$\boxed{\lim_{x \to 0^+} x \ln x = 0}.$$

Q12. Evaluate the limit: $\lim_{x \to \infty} x^3 e^{-x^2}$.

S12

- Rewrite:
$$x^3 e^{-x^2} = \frac{x^3}{e^{x^2}}.$$
As $x \to \infty$, both numerator and denominator tend to ∞, so we have an $\dfrac{\infty}{\infty}$ form.
- **1st L'Hôpital:**
$$\lim_{x \to \infty} \frac{x^3}{e^{x^2}} = \lim_{x \to \infty} \frac{3x^2}{2x e^{x^2}} = \lim_{x \to \infty} \frac{3x}{2e^{x^2}},$$
still $\dfrac{\infty}{\infty}$.
- **2nd L'Hôpital:**
$$\lim_{x \to \infty} \frac{3x}{2e^{x^2}} = \lim_{x \to \infty} \frac{3}{4x e^{x^2}},$$
now the denominator $\to \infty$.

Thus the limit is
$$\boxed{\lim_{x \to \infty} x^3 e^{-x^2} = 0}.$$

Q13. Evaluate the limit: $\lim_{x \to \infty} (1 + 2x)^{1/(2 \ln x)}$.

S13

1. Since the expression is positive, set
$$y = (1 + 2x)^{1/(2 \ln x)}.$$
Take natural logarithms:
$$\ln y = \frac{\ln(1 + 2x)}{2 \ln x}.$$
2. The limit
$$L = \lim_{x \to \infty} \ln y = \lim_{x \to \infty} \frac{\ln(1 + 2x)}{2 \ln x}$$
is an indeterminate form of type $\frac{\infty}{\infty}$, so L'Hôpital's Rule applies.

3. Differentiate the numerator and denominator:
$$\frac{d}{dx}\ln(1+2x) = \frac{2}{1+2x}, \qquad \frac{d}{dx}(2\ln x) = \frac{2}{x}.$$
4. Apply L'Hôpital's Rule:
$$L = \lim_{x\to\infty} \frac{\frac{2}{1+2x}}{\frac{2}{x}} = \lim_{x\to\infty} \frac{2}{1+2x}\cdot\frac{x}{2} = \lim_{x\to\infty} \frac{x}{1+2x}.$$
5. Evaluate this limit:
$$\lim_{x\to\infty} \frac{x}{1+2x} = \lim_{x\to\infty} \frac{1}{2+\frac{1}{x}} = \frac{1}{2}.$$
6. Therefore,
$$\lim_{x\to\infty} \ln y = \frac{1}{2} \qquad \Rightarrow \qquad \lim_{x\to\infty} y = e^{1/2}.$$
$$\boxed{\lim_{x\to\infty} (1+2x)^{1/(2\ln x)} = \sqrt{e}.}$$

Q14. Evaluate the limit: $\lim_{x\to 1^+} \dfrac{\ln x}{\sin(\pi x)}$.

S14

1. As $x \to 1^+$,
$$\ln x \to 0, \qquad \sin(\pi x) \to \sin(\pi) = 0,$$
so the expression is of the indeterminate form $\frac{0}{0}$ and L'Hôpital's Rule applies.
2. Differentiate the numerator and denominator:
$$\frac{d}{dx}(\ln x) = \frac{1}{x}, \qquad \frac{d}{dx}(\sin(\pi x)) = \pi\cos(\pi x).$$
3. Apply L'Hôpital's Rule:
$$\lim_{x\to 1^+} \frac{\ln x}{\sin(\pi x)} = \lim_{x\to 1^+} \frac{\frac{1}{x}}{\pi\cos(\pi x)} = \frac{1}{\pi\cos(\pi)} = \frac{1}{\pi(-1)} = -\frac{1}{\pi}.$$
$$\boxed{\lim_{x\to 1^+} \frac{\ln x}{\sin(\pi x)} = -\frac{1}{\pi}}$$

Q15. Evaluate the limit: $\lim_{x\to 0} \dfrac{2x - \tan^{-1}(2x)}{x^3}$.

S15

- **IF:** At $x = 0$, numerator $2x - \tan^{-1}(2x) \to 0 - 0 = 0$ and denominator $x^3 \to 0$, so we have $\dfrac{0}{0}$.
- **Apply L'Hôpital's Rule once.** Let $N(x) = 2x - \tan^{-1}(2x)$, $D(x) = x^3$.
$$N'(x) = 2 - \frac{2}{1+4x^2} = 2\left(1 - \frac{1}{1+4x^2}\right) = \frac{8x^2}{1+4x^2}, \qquad D'(x) = 3x^2.$$
Then
$$\lim_{x\to 0} \frac{2x - \tan^{-1}(2x)}{x^3} = \lim_{x\to 0} \frac{N'(x)}{D'(x)} = \lim_{x\to 0} \frac{\frac{8x^2}{1+4x^2}}{3x^2} = \lim_{x\to 0} \frac{8}{3(1+4x^2)} = \frac{8}{3}.$$

Therefore,
$$\boxed{\lim_{x\to 0} \frac{2x - \tan^{-1}(2x)}{x^3} = \frac{8}{3}}.$$

Q16. Evaluate the limit: $\lim_{x\to 0^+} x\ln x$

S16

1. **Initial Indeterminate Form (IF):** Substituting $x = 0^+$ yields $0 \cdot (-\infty)$ [1].
2. **Convert to $\frac{\infty}{\infty}$ Form:** Rewrite the product as a quotient suitable for L'Hôpital's Rule:

$$\lim_{x \to 0^+} \frac{\ln x}{1/x}$$

3. **Apply L'Hôpital's Rule:** Differentiate the numerator and the denominator:

$$\lim_{x \to 0^+} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(1/x)} = \lim_{x \to 0^+} \frac{1/x}{-1/x^2}$$

4. **Simplify and Evaluate:**

$$\lim_{x \to 0^+} \left(\frac{1}{x} \cdot (-x^2) \right) = \lim_{x \to 0^+} (-x) = \mathbf{0}$$

Q17. Evaluate the limit: $\lim\limits_{x \to \infty} x^3 e^{-x^2}$.

S17

1. Rewrite the expression as a quotient suitable for L'Hôpital's Rule:

$$x^3 e^{-x^2} = \frac{x^3}{e^{x^2}}.$$

   As $x \to \infty$, both numerator and denominator $\to \infty$, so we have an ∞/∞ indeterminate form.
2. Apply L'Hôpital's Rule once:

$$\lim_{x \to \infty} \frac{x^3}{e^{x^2}} = \lim_{x \to \infty} \frac{3x^2}{2xe^{x^2}} = \lim_{x \to \infty} \frac{3x}{2e^{x^2}},$$

   which is still of the type ∞/∞.
3. Apply L'Hôpital's Rule a second time:

$$\lim_{x \to \infty} \frac{3x}{2e^{x^2}} = \lim_{x \to \infty} \frac{3}{4xe^{x^2}} = 0,$$

   since the denominator $4xe^{x^2} \to \infty$.

Thus

$$\boxed{\lim_{x \to \infty} x^3 e^{-x^2} = 0.}$$

Q18. Evaluate the limit: $\lim\limits_{x \to \infty} (1 + 2x)^{1/(2 \ln x)}$.

S18

1. Since the base is positive, set

$$y = (1 + 2x)^{1/(2 \ln x)}, \qquad y > 0.$$

   Take natural logs:

$$\ln y = \frac{\ln(1 + 2x)}{2 \ln x}.$$

   We first compute $L = \lim\limits_{x \to \infty} \ln y$.
2. As $x \to \infty$, both $\ln(1+2x)$ and $2 \ln x$ go to ∞, giving an ∞/∞ form, so we apply L'Hôpital's Rule:

$$L = \lim_{x \to \infty} \frac{\ln(1 + 2x)}{2 \ln x} = \lim_{x \to \infty} \frac{\dfrac{2}{1 + 2x}}{\dfrac{2}{x}} = \lim_{x \to \infty} \frac{2}{1 + 2x} \cdot \frac{x}{2} = \lim_{x \to \infty} \frac{x}{1 + 2x} = \frac{1}{2}.$$

3. Therefore,

$$\lim_{x\to\infty} \ln y = \frac{1}{2} \quad \Longrightarrow \quad \lim_{x\to\infty} y = e^{1/2} = \sqrt{e}.$$

Hence

$$\boxed{\lim_{x\to\infty} (1+2x)^{1/(2\ln x)} = \sqrt{e}.}$$

Q19. Evaluate the limit: $\lim_{x\to 1^+} (\ln x)\sin(\pi x)$.

S19

1. Both functions are continuous at $x = 1$ (from the right):

$$\lim_{x\to 1^+} \ln x = \ln 1 = 0, \qquad \lim_{x\to 1^+} \sin(\pi x) = \sin(\pi \cdot 1) = 0.$$

2. Since this is just the product of two functions with finite limits, and $0 \cdot 0$ is *not* an indeterminate form, we can take the product of the limits:

$$\lim_{x\to 1^+} (\ln x)\sin(\pi x) = \left(\lim_{x\to 1^+} \ln x\right)\left(\lim_{x\to 1^+} \sin(\pi x)\right) = 0 \cdot 0 = 0.$$

Thus

$$\boxed{\lim_{x\to 1^+} (\ln x)\sin(\pi x) = 0.}$$

Q20. Evaluate the limit: $\lim_{x\to 0} \dfrac{2x - \tan^{-1}(2x)}{x^3}$.

S20

1. Direct substitution gives

$$2x - \tan^{-1}(2x) \xrightarrow[x\to 0]{} 0, \qquad x^3 \xrightarrow[x\to 0]{} 0,$$

so the expression is of the indeterminate form $\dfrac{0}{0}$.

2. Apply L'Hôpital's Rule once:

$$\lim_{x\to 0} \frac{2x - \tan^{-1}(2x)}{x^3} = \lim_{x\to 0} \frac{\dfrac{d}{dx}\big(2x - \tan^{-1}(2x)\big)}{\dfrac{d}{dx}(x^3)}.$$

Compute derivatives:

$$\frac{d}{dx}(2x) = 2, \qquad \frac{d}{dx}\big(\tan^{-1}(2x)\big) = \frac{2}{1+(2x)^2} = \frac{2}{1+4x^2},$$

so

$$\frac{d}{dx}\big(2x - \tan^{-1}(2x)\big) = 2 - \frac{2}{1+4x^2} = \frac{2(1+4x^2)-2}{1+4x^2} = \frac{8x^2}{1+4x^2},$$

and

$$\frac{d}{dx}(x^3) = 3x^2.$$

Therefore

$$\lim_{x\to 0} \frac{2x - \tan^{-1}(2x)}{x^3} = \lim_{x\to 0} \frac{\dfrac{8x^2}{1+4x^2}}{3x^2} = \lim_{x\to 0} \frac{8}{3(1+4x^2)} = \frac{8}{3}.$$

Hence

$$\boxed{\lim_{x\to 0} \frac{2x - \tan^{-1}(2x)}{x^3} = \frac{8}{3}.}$$

4. Comprehensive Curve Sketching

Q21. Find the intervals of concavity for $\mathbf{f(x) = x^3 + 6x^2 + 9x}$.

S21

1. **Find** $f''(x)$: $f'(x) = 3x^2 + 12x + 9$. Thus $f''(x) = 6x + 12$. 2. **Inflection point candidate**: $f''(x) = 0$ gives $x = -2$. 3. **Concavity**: The graph is concave down on $(-\infty, \mathbf{-2})$ (since $f'' < 0$) and concave up on $(\mathbf{-2}, \infty)$ (since $f'' > 0$).

Q22. Find the horizontal and vertical asymptotes for $\mathbf{y = \dfrac{x^2}{x^2 + 9}}$.

S22

This analysis uses the initial steps of curve sketching focusing on domain and asymptotic behavior.

A. **DOMAIN:** $\mathbf{D} = (-\infty, \infty)$ (denominator $x^2 + 9$ is always positive).

B. **INTERCEPTS:** $(0, 0)$.

D. **ASYMPTOTES:**

- Vertical asymptotes: none.
- Horizontal asymptotes: $\lim_{x \to \pm\infty} \dfrac{x^2}{x^2 + 9} = 1$. Hence $\mathbf{y = 1}$.

Q23. Find the horizontal and vertical asymptotes for $\mathbf{y = \dfrac{x}{x^2 - 9}}$.

S23

A. **DOMAIN:** $\mathbf{x \neq \pm 3}$.

B. **INTERCEPTS:** $(0, 0)$.

D. **ASYMPTOTES:**

- Vertical asymptotes: $x^2 - 9 = 0 \Rightarrow x = \pm 3$.
- Horizontal asymptote: degree of numerator $<$ degree of denominator, so $\lim_{x \to \pm\infty} y = 0$. Hence $\mathbf{y = 0}$.

Q24. Is the function $\mathbf{y = \dfrac{x^2}{x^2 + 9}}$ even or odd?

S24

Since $f(-x) = f(x)$, the function is even (y-axis symmetry).

Q25. Find the vertical asymptote for $\mathbf{y = \dfrac{x}{x - 5}}$.

S25

The vertical asymptote occurs where the denominator is zero and the numerator is non-zero. Thus the vertical asymptote is at $\mathbf{x = 5}$.

Q26. Find the vertical asymptote for $\mathbf{y = \dfrac{x^3 + 1}{x^3 + x}}$.

S26

Since $x^3 + x = x(x^2 + 1)$, the only real root of the denominator is $x = 0$. Thus the vertical asymptote is at $\mathbf{x = 0}$.

Q27. Use curve sketching guidelines to analyze $\mathbf{y = x^3 + 6x^2 + 9x}$.

S27

A. **DOMAIN:** $\mathbf{D} = (-\infty, \infty)$.

B. **INTERCEPTS:** $x(x+3)^2 = 0$. Thus the intercepts are $(\mathbf{0}, \mathbf{0})$ and $(\mathbf{-3}, \mathbf{0})$.

C. **SYMMETRY:** No symmetry (neither even nor odd).

D. **ASYMPTOTES:** None.

E. **INCREASE/DECREASE (f'):** $f'(x) = 3(x+3)(x+1)$. Increasing on $(-\infty, \mathbf{-3}) \cup (\mathbf{-1}, \infty)$; decreasing on $(\mathbf{-3}, \mathbf{-1})$.

F. **LOCAL EXTREMA:** Local maximum at $(\mathbf{-3}, \mathbf{0})$; local minimum at $(\mathbf{-1}, \mathbf{-4})$.

G. **CONCAVITY / INFLECTION (f''):** $f''(x) = 6x + 12$. The graph is concave down on $(-\infty, \mathbf{-2})$ and concave up on $(\mathbf{-2}, \infty)$. There is an inflection point at $(\mathbf{-2}, \mathbf{-2})$.

H. **SKETCH TABLE:**

x	$(-\infty,-3)$	-3	$(-3,-2)$	-2	$(-2,-1)$	-1	$(-1,\infty)$
$f'(x)$	$+$	0	$-$	$-$	$-$	0	$+$
$f''(x)$	$-$	$-$	$-$	0	$+$	$+$	$+$
y-behavior	$\nearrow$	max	$\searrow$	$\searrow$	$\searrow$	min	$\nearrow$
Concavity	$\frown$	$\frown$	$\frown$	IP	$\smile$	$\smile$	$\smile$

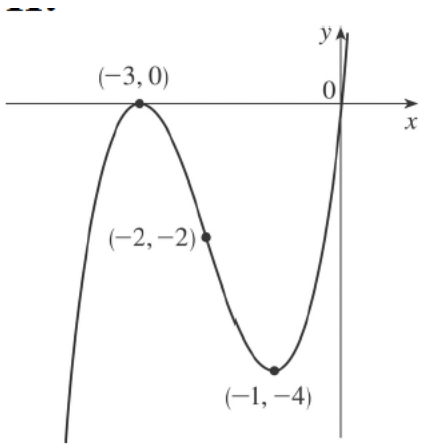

Figure 1: Sketch of $y = x^3 + 6x^2 + 9x$.

Q28. Use curve sketching guidelines to analyze $\mathbf{f(x) = 8x^2 - x^4}$.

S28

A. **DOMAIN:** $\mathbf{D} = (-\infty, \infty)$.

B. **INTERCEPTS:** $x^2(8 - x^2) = 0$. Thus the intercepts are $(\mathbf{0}, \mathbf{0})$ and $(\pm\mathbf{2\sqrt{2}}, \mathbf{0})$.

C. **SYMMETRY:** $f(-x) = f(x)$, so f is even (y-axis symmetry).

D. **ASYMPTOTES:** None.

E. **INCREASE/DECREASE (f'):** $f'(x) = 16x - 4x^3 = -4x(x-2)(x+2)$. The function is increasing on $(-\infty, \mathbf{-2}) \cup (\mathbf{0}, \mathbf{2})$ and decreasing on $(\mathbf{-2}, \mathbf{0}) \cup (\mathbf{2}, \infty)$.

F. **LOCAL EXTREMA:** Local maxima at $(\mathbf{-2}, \mathbf{16})$ and $(\mathbf{2}, \mathbf{16})$; local minimum at $(\mathbf{0}, \mathbf{0})$.

G. **CONCAVITY / INFLECTION (f''):** $f''(x) = 16-12x^2 = 4(4-3x^2)$. Setting $f''(x) = 0$ gives $x^2 = \frac{4}{3}$, so inflection points occur at $x = \pm\frac{2}{\sqrt{3}}$. The graph is concave down on $(-\infty, \mathbf{-2/\sqrt{3}}) \cup (\mathbf{2/\sqrt{3}}, \infty)$ and concave up on $(\mathbf{-2/\sqrt{3}}, \mathbf{2/\sqrt{3}})$.

H. **SKETCH TABLE:**

x	$(-\infty,-2)$	-2	$(-2,-2/\sqrt{3})$	$-2/\sqrt{3}$	$(-2/\sqrt{3},0)$	0	$(0,2/\sqrt{3})$	$2/\sqrt{3}$	$(2/\sqrt{3},2)$	2	(
$f'(x)$	$+$	0	$-$	$-$	$-$	0	$+$	$+$	$+$	0	
$f''(x)$	$-$	$-$	$-$	0	$+$	$+$	$+$	0	$-$	$-$	
y-behavior	$\nearrow$	max	$\searrow$	$\searrow$	$\searrow$	min	$\nearrow$	$\nearrow$	$\nearrow$	max	
Concavity	$\frown$	$\frown$	$\frown$	IP	$\smile$	$\smile$	$\smile$	IP	$\frown$	$\frown$	

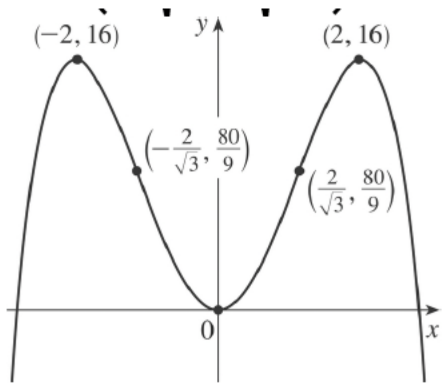

Figure 2: Sketch of $y = 8x^2 - x^4$.

Q29. Analyze the critical numbers for $\mathbf{f(x) = x^4 + cx^2}$. Identify the transitional value of c.

S29

1. **Critical points**: $f'(x) = 4x^3 + 2cx = 2x(2x^2 + c)$. 2. **Analysis**: Critical points exist at $x = 0$ and where $x^2 = -c/2$.

- If $c \geq 0$, there is only one real critical point ($x = 0$).
- If $c < 0$, there are three real critical points.

The transitional value where the nature/number of critical points changes is $\mathbf{c = 0}$.

Q30. Use curve sketching guidelines to analyze $\mathbf{y = x/(x-1)}$.

S30

A. **DOMAIN:** $\mathbf{x \neq 1}$.

B. **INTERCEPTS:** $(\mathbf{0}, \mathbf{0})$.

C. **SYMMETRY:** No symmetry.

D. **ASYMPTOTES:** Horizontal asymptote $\mathbf{y = 1}$; vertical asymptote $\mathbf{x = 1}$.

E. **INCREASE/DECREASE (f'):** $f'(x) = -\dfrac{1}{(x-1)^2}$, always negative on the domain, so f is always decreasing.

F. **LOCAL EXTREMA:** None.

G. **CONCAVITY / INFLECTION (f''):** $f''(x) = \frac{2}{(x-1)^3}$. Concave down on $(-\infty, \mathbf{1})$ and concave up on $(\mathbf{1}, \infty)$. There is no inflection point because $x = 1$ is not in the domain.

H. **SKETCH TABLE:**

x	$(-\infty, 1)$	1	$(1, \infty)$
$f'(x)$	$-$	undef	$-$
$f''(x)$	$-$	undef	$+$
y-behavior	$\searrow$	VA	$\searrow$
Concavity	$\frown$		$\smile$

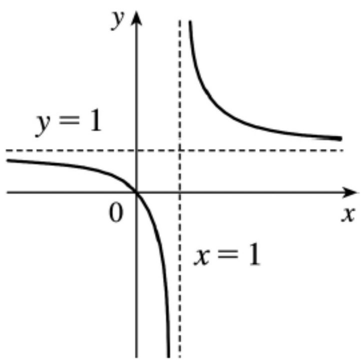

Figure 3: Sketch of $y = x/(x-1)$.

Q31. Use curve sketching guidelines to analyze $\mathbf{y = 1/(x^2 - 9)}$.

S31

A. **DOMAIN:** $\mathbf{x \neq \pm 3}$.

B. **INTERCEPTS:** y-intercept $(\mathbf{0}, \mathbf{-1/9})$; no x-intercept.

C. **SYMMETRY:** Even.

D. **ASYMPTOTES:** Horizontal asymptote $\mathbf{y = 0}$; vertical asymptotes $\mathbf{x = \pm 3}$.

E. **INCREASE/DECREASE (f'):** $f'(x) = -\frac{2x}{(x^2-9)^2}$. Increasing on $(-\infty, \mathbf{-3}) \cup (\mathbf{-3}, \mathbf{0})$; decreasing on $(\mathbf{0}, \mathbf{3}) \cup (\mathbf{3}, \infty)$.

F. **LOCAL EXTREMA:** Local maximum at $(\mathbf{0}, \mathbf{-1/9})$.

G. **CONCAVITY / INFLECTION (f''):** $f''(x) = \frac{6x^2 + 18}{(x^2-9)^3}$. Concave up on $(-\infty, \mathbf{-3}) \cup (\mathbf{3}, \infty)$ and concave down on $(\mathbf{-3}, \mathbf{3})$. No inflection point.

H. **SKETCH TABLE:**

x	$(-\infty, -3)$	-3	$(-3, 0)$	0	$(0, 3)$	3	$(3, \infty)$
$f'(x)$	$+$	undef	$+$	0	$-$	undef	$-$
$f''(x)$	$+$	undef	$-$	$-$	$-$	undef	$+$
y-behavior	$\nearrow$	VA	$\nearrow$	max	$\searrow$	VA	$\searrow$
Concavity	$\smile$		$\frown$	$\frown$	$\frown$		$\smile$

Q32. Use curve sketching guidelines to analyze $\mathbf{y = x^4 + 4x^3}$.

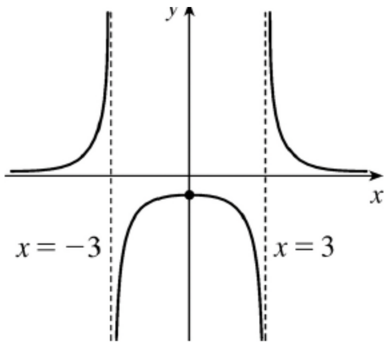

Figure 4: Sketch of $y = 1/(x^2 - 9)$.

S32

A. **DOMAIN:** $\mathbf{D} = (-\infty, \infty)$.

B. **INTERCEPTS:** $x^3(x+4) = 0$. Thus the intercepts are $(\mathbf{0}, \mathbf{0})$ and $(-\mathbf{4}, \mathbf{0})$.

C. **SYMMETRY:** No symmetry (neither even nor odd).

D. **ASYMPTOTES:** None.

E. **INCREASE/DECREASE (f'):**

$$f'(x) = 4x^3 + 12x^2 = 4x^2(x+3).$$

Since $4x^2 \geq 0$, the sign of $f'(x)$ is determined by $(x+3)$ except at $x = 0$:

- Decreasing on $(-\infty, -\mathbf{3})$,
- Increasing on $(-\mathbf{3}, \mathbf{0})$ and on $(\mathbf{0}, \infty)$.

F. **LOCAL EXTREMA:** Critical points: $x = -3, 0$. $f(-3) = (-3)^4 + 4(-3)^3 = 81 - 108 = -27$ and $f''(-3) = 12(-3)(-1) = 36 > 0$, so there is a local minimum at $(-\mathbf{3}, -\mathbf{27})$. At $x = 0$, $f'(0) = 0$ but f' is positive on both sides, so there is no local extremum at $x = 0$.

G. **CONCAVITY / INFLECTION (f''):**

$$f''(x) = 12x^2 + 24x = 12x(x+2).$$

Hence:

- Concave up on $(-\infty, -\mathbf{2})$ and on $(\mathbf{0}, \infty)$,
- Concave down on $(-\mathbf{2}, \mathbf{0})$.

Since the concavity changes at $x = -2$ and at $x = 0$, there are inflection points at

$$f(-2) = 16 + 4(-8) = -16 \quad \Rightarrow \quad (-2, -16), \qquad f(0) = 0 \quad \Rightarrow \quad (0, 0).$$

The point $(0, 0)$ is an inflection point with a horizontal tangent.

H. **SKETCH TABLE:**

x	$(-\infty, -3)$	-3	$(-3, -2)$	-2	$(-2, 0)$	0	$(0, \infty)$
$f'(x)$	$-$	0	$+$	$+$	$+$	0	$+$
$f''(x)$	$+$	$+$	$+$	0	$-$	0	$+$
y-behavior	$\searrow$	min	$\nearrow$	$\nearrow$	$\nearrow$		$\nearrow$
Concavity	$\smile$	$\smile$	$\smile$	IP	$\frown$	IP	$\smile$

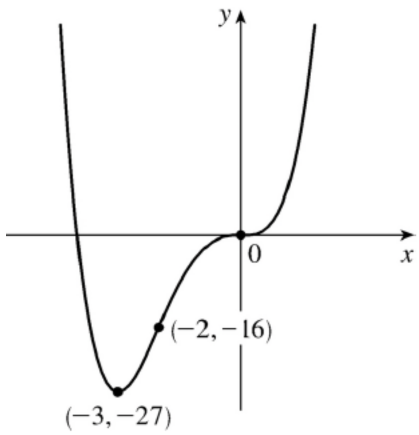

Figure 5: Sketch of $y = x^4 + 4x^3$.

4. Linear Approximations and Differentials

Find the linearization $L(x)$ of the function $f(x)$ at the specified value of a.
Q33. $f(x) = x^3$, $a = 1$.

S33

1. Calculate the function value at a:
$$f(1) = 1^3 = 1$$
2. Calculate the derivative: $f'(x) = 3x^2$.
3. Evaluate the derivative at a:
$$f'(1) = 3(1)^2 = 3$$
4. Apply the linearization formula $L(x) = f(a) + f'(a)(x - a)$ [1]:
$$L(x) = 1 + 3(x - 1) = \mathbf{3x - 2}$$

Q34. $f(x) = \ln x$, $a = 1$.

S34

1. Calculate the function value at a:
$$f(1) = \ln(1) = 0$$
2. Calculate the derivative: $f'(x) = 1/x$.
3. Evaluate the derivative at a:
$$f'(1) = 1/1 = 1$$
4. Apply the linearization formula:
$$L(x) = 0 + 1(x - 1) = \mathbf{x - 1}$$

Q35. $f(x) = \cos x$, $a = \pi/2$.

S35

1. Calculate the function value at a:

$$f(\pi/2) = \cos(\pi/2) = 0$$

2. Calculate the derivative: $f'(x) = -\sin x$.

3. Evaluate the derivative at a:

$$f'(\pi/2) = -\sin(\pi/2) = -1$$

4. Apply the linearization formula :

$$L(x) = 0 + (-1)(x - \pi/2) = -\mathbf{x} + \frac{\pi}{\mathbf{2}}$$

Q36. $f(x) = \sqrt[3]{x} = x^{1/3}$, $a = -8$.

S36

1. Calculate the function value at a:

$$f(-8) = (-8)^{1/3} = -2$$

2. Calculate the derivative: $f'(x) = \frac{1}{3}x^{-2/3}$.

3. Evaluate the derivative at a:

$$f'(-8) = \frac{1}{3}(-8)^{-2/3} = \frac{1}{3}\left(\frac{1}{(-8)^{2/3}}\right) = \frac{1}{3}\left(\frac{1}{4}\right) = \frac{1}{12}$$

4. Apply the linearization formula :

$$L(x) = -2 + \frac{1}{12}(x+8) = \frac{1}{12}x + \frac{8}{12} - 2 = \frac{\mathbf{1}}{\mathbf{12}}\mathbf{x} - \frac{\mathbf{4}}{\mathbf{3}}$$

Use differentials (or, equivalently, a linear approximation) to estimate the given number [4].

Q37. Estimate $\sqrt{99.8}$.

S37

1. Define $f(x) = \sqrt{x}$. Choose $a = 100$ and $\Delta x = -0.2$.
2. Calculate $f(a)$: $f(100) = \sqrt{100} = 10$.
3. Find the derivative: $f'(x) = \frac{1}{2\sqrt{x}}$.
4. Calculate $f'(a)$: $f'(100) = \frac{1}{2\sqrt{100}} = \frac{1}{20} = 0.05$.
5. Apply the differential approximation $f(a + \Delta x) \approx f(a) + f'(a)\Delta x$:

$$\sqrt{99.8} \approx 10 + (0.05)(-0.2) = 10 - 0.01 = \mathbf{9.99}$$

Q38. Estimate $(1.01)^6$.

S38

1. Define $f(x) = x^6$. Choose $a = 1$ and $\Delta x = 0.01$.
2. Calculate $f(a)$: $f(1) = 1^6 = 1$.
3. Find the derivative: $f'(x) = 6x^5$.
4. Calculate $f'(a)$: $f'(1) = 6(1)^5 = 6$.
5. Apply the differential approximation:

$$(1.01)^6 \approx 1 + 6(0.01) = 1 + 0.06 = \mathbf{1.06}$$

Q39. Estimate $\sin 59°$.

S39

1. Define $f(x) = \sin x$. Choose $a = 60° = \pi/3$ (in radians) and $\Delta x = -1° = -\pi/180$ (in radians).
2. Calculate $f(a)$: $f(\pi/3) = \sin(\pi/3) = \sqrt{3}/2$.
3. Find the derivative: $f'(x) = \cos x$.
4. Calculate $f'(a)$: $f'(\pi/3) = \cos(\pi/3) = 1/2$.
5. Apply the differential approximation:

$$\sin 59° \approx \frac{\sqrt{3}}{2} + \frac{1}{2}\left(-\frac{\pi}{180}\right) \approx \mathbf{0.8573}$$

(Using $\pi/180 \approx 0.01745$ and $\sqrt{3}/2 \approx 0.8660$)

Q40. Estimate $(1.02)^{-3}$.

S40

1. Define $f(x) = x^{-3}$. Choose $a = 1$ and $\Delta x = 0.02$.
2. Calculate $f(a)$: $f(1) = 1^{-3} = 1$.
3. Find the derivative: $f'(x) = -3x^{-4}$.
4. Calculate $f'(a)$: $f'(1) = -3(1)^{-4} = -3$.
5. Apply the differential approximation:

$$(1.02)^{-3} \approx 1 + (-3)(0.02) = 1 - 0.06 = \mathbf{0.94}$$

Q41. Estimate $\dfrac{1}{4.001}$.

S41

1. Define $f(x) = 1/x = x^{-1}$. Choose $a = 4$ and $\Delta x = 0.001$.
2. Calculate $f(a)$: $f(4) = 1/4 = 0.25$.
3. Find the derivative: $f'(x) = -x^{-2} = -1/x^2$.
4. Calculate $f'(a)$: $f'(4) = -1/16 = -0.0625$.
5. Apply the differential approximation:

$$\frac{1}{4.001} \approx 0.25 + (-0.0625)(0.001) = 0.25 - 0.0000625 = \mathbf{0.2499375}$$

Q42. Estimate $\arctan(1.02)$.

S42

1. Define $f(x) = \arctan x$. Choose $a = 1$ and $\Delta x = 0.02$.
2. Calculate $f(a)$: $f(1) = \arctan(1) = \pi/4$.
3. Find the derivative: $f'(x) = \frac{1}{1+x^2}$.
4. Calculate $f'(a)$: $f'(1) = \frac{1}{1+1^2} = 1/2 = 0.5$.
5. Apply the differential approximation:

$$\arctan(1.02) \approx \frac{\pi}{4} + (0.5)(0.02) = \frac{\pi}{4} + 0.01$$

(Numeric approximation: $\approx 0.785398 + 0.01 = \mathbf{0.7954}$)

5. Related Rates Problems

Q43. Volume of a Sphere If air is being pumped into a spherical balloon so that its radius r increases at a constant rate of 2 cm/s, how fast is the volume V of the balloon increasing when the radius is $r = 5$ cm?

S43

1. Relationship: The volume of a sphere is $V = \frac{4}{3}\pi r^3$.
2. Given rates: $\frac{dr}{dt} = 2$ cm/s (constant). We seek $\frac{dV}{dt}$ when $r = 5$.
3. Differentiate implicitly with respect to time t (Chain Rule) :

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

4. Substitute values:

$$\frac{dV}{dt} = 4\pi(5)^2(2) = 4\pi(25)(2) = \mathbf{200}\pi \text{ cm}^{\mathbf{3}}/\text{s}$$

Q44. Area of an Oil Spill Suppose oil spills from a ruptured tanker and spreads in a circular pattern. If the radius r of the oil spill increases at a constant rate of 1 m/s, how fast is the area A of the spill increasing when the radius is $r = 30$ m?

S44

1. Relationship: The area of a circle is $A = \pi r^2$.
2. Given rates: $\frac{dr}{dt} = 1$ m/s. We seek $\frac{dA}{dt}$ when $r = 30$.
3. Differentiate implicitly:

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

4. Substitute values:

$$\frac{dA}{dt} = 2\pi(30)(1) = \mathbf{60}\pi \text{ m}^{\mathbf{2}}/\text{s}$$

Q45. Pythagorean Relationship If $z^2 = x^2 + y^2$, and $\frac{dx}{dt} = 2$ and $\frac{dy}{dt} = 3$, find $\frac{dz}{dt}$ when $x = 5$ and $y = 12$.

S45

1. Find z at the given instant:
$$z = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = 13.$$
2. Differentiate the relationship implicitly:
$$2z\frac{dz}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$$
3. Solve for the unknown rate:
$$\frac{dz}{dt} = \frac{x\frac{dx}{dt} + y\frac{dy}{dt}}{z}$$
4. Substitute values:
$$\frac{dz}{dt} = \frac{(5)(2) + (12)(3)}{13} = \frac{10 + 36}{13} = \frac{46}{13} \approx \mathbf{3.538}$$

Q46. Cube Volume If the volume of a cube is $V = x^3$ and $\frac{dV}{dt} = 10$ cm^3/min, find $\frac{dx}{dt}$ when $x = 3$ cm.

S46

1. Relationship: $V = x^3$.
2. Differentiate implicitly:
$$\frac{dV}{dt} = 3x^2\frac{dx}{dt}$$
3. Solve for $\frac{dx}{dt}$:
$$\frac{dx}{dt} = \frac{1}{3x^2}\frac{dV}{dt}$$
4. Substitute values:
$$\frac{dx}{dt} = \frac{1}{3(3)^2}(10) = \frac{10}{27} \approx \mathbf{0.370} \text{ cm/min}$$

Q47. Variable Relationship If $y = x^2 + 3$, find $\frac{dy}{dt}$ when $x = 1$, given that $\frac{dx}{dt} = 2$.

S47

1. Relationship: $y = x^2 + 3$.
2. Differentiate implicitly:
$$\frac{dy}{dt} = 2x\frac{dx}{dt}$$
3. Substitute values ($x = 1$, $\frac{dx}{dt} = 2$):
$$\frac{dy}{dt} = 2(1)(2) = \mathbf{4}$$

Q48. Particle along a Curve A particle moves along the curve $y = \sqrt{x}$. As it reaches the point $(4, 2)$, the x-coordinate is increasing at a rate of 3 cm/s. How fast is the y-coordinate of the point changing at that instant?

S48

1. Relationship: $y = x^{1/2}$.
2. Given rates: $\frac{dx}{dt} = 3$ cm/s. We seek $\frac{dy}{dt}$ when $x = 4$.
3. Differentiate implicitly:
$$\frac{dy}{dt} = \frac{1}{2}x^{-1/2}\frac{dx}{dt}$$
4. Substitute values:
$$\frac{dy}{dt} = \frac{1}{2\sqrt{4}}(3) = \frac{1}{4}(3) = \mathbf{0.75}\text{ cm/s}$$

Q49. Shadow Length (Similar Triangles) A man 2 m tall walks away from a 12 m tall lamppost at a speed of 1.6 m/s. How fast is the tip of his shadow moving when he is 8 m from the pole? [9]

S49

1. Define variables: Let x be the distance from the man to the pole, and s be the length of his shadow. Let $z = x + s$ be the distance from the pole to the tip of the shadow. We seek $\frac{dz}{dt}$. We are given $\frac{dx}{dt} = 1.6$ m/s.
2. Geometric relationship (Similar Triangles): $\frac{12}{2} = \frac{x+s}{s}$.
$$6s = x + s \implies 5s = x \implies s = \frac{1}{5}x$$
3. Relationship for z: $z = x + s = x + \frac{1}{5}x = \frac{6}{5}x$.
4. Differentiate implicitly:
$$\frac{dz}{dt} = \frac{6}{5}\frac{dx}{dt}$$
5. Substitute known rate:
$$\frac{dz}{dt} = \frac{6}{5}(1.6) = \mathbf{1.92}\text{ m/s}$$
(Note: The rate is constant and independent of x.)

Q50. Conical Tank (Volume Rate) A water tank in the shape of an inverted circular cone has a base radius of 4 m and a height of 12 m. If water is poured into the tank at a rate of 2 m^3/min, how fast is the water level h rising when the water is 6 m deep? (Adapted from similar problems like Q19 in source .)

S50

1. Relationship: $V = \frac{1}{3}\pi r^2 h$.
2. Eliminate r using similar triangles: $\frac{r}{h} = \frac{4}{12} \implies r = \frac{1}{3}h$.
3. Substitute r into volume formula:
$$V = \frac{1}{3}\pi\left(\frac{1}{3}h\right)^2 h = \frac{\pi}{27}h^3.$$
4. Given rates: $\frac{dV}{dt} = 2$ m^3/min. We seek $\frac{dh}{dt}$ when $h = 6$.
5. Differentiate implicitly:
$$\frac{dV}{dt} = \frac{\pi}{27}(3h^2)\frac{dh}{dt} = \frac{\pi h^2}{9}\frac{dh}{dt}$$

6. Solve for $\frac{dh}{dt}$:
$$\frac{dh}{dt} = \frac{9}{\pi h^2}\frac{dV}{dt}$$
7. Substitute values:
$$\frac{dh}{dt} = \frac{9}{\pi(6)^2}(2) = \frac{18}{36\pi} = \frac{\mathbf{1}}{\mathbf{2}\boldsymbol{\pi}} \text{ m/min}$$

Q51. Trough Filling (Isosceles Cross-Section) A trough is 10 ft long and its ends are isosceles triangles that are 3 ft across at the top and 1 ft high. If the trough is being filled with water at a rate of 12 ft^3/min, how fast is the water level rising when the water is 6 inches (0.5 ft) deep?

S51

1. Relationship: The volume V is the area of the triangular cross-section ($A = \frac{1}{2}bh$) times the length ($L = 10$ ft), so $V = 5bh$.
2. Eliminate b using similar triangles: $\frac{b}{h} = \frac{3}{1} \implies b = 3h$.
3. Substitute b into volume formula:
$$V = 5(3h)h = 15h^2.$$
4. Given rates: $\frac{dV}{dt} = 12$ ft^3/min. We seek $\frac{dh}{dt}$ when $h = 0.5$ ft.
5. Differentiate implicitly:
$$\frac{dV}{dt} = 30h\frac{dh}{dt}$$
6. Solve for $\frac{dh}{dt}$:
$$\frac{dh}{dt} = \frac{1}{30h}\frac{dV}{dt}$$
7. Substitute values:
$$\frac{dh}{dt} = \frac{1}{30(0.5)}(12) = \frac{12}{15} = \mathbf{0.8} \text{ ft/min}$$

Q52. Distance Between Ships (Law of Cosines) Two ships leave port at 12:00 noon. Ship A travels west at 35 mi/h and Ship B travels 4 hours later at 25 mi/h along a path 120° clockwise from the West direction (assuming 120° relative to Ship A's path for simplicity, leading to the necessary **60°** angle between paths, as implied by the Law of Cosines solution structure found in related sources). Find the rate at which the distance between the ships is changing at $t = 4$ hours (Ship A has traveled 4 hours, Ship B has traveled 4 hours).

S52

1. Define variables: Let x be the distance of Ship A and y be the distance of Ship B. z is the distance between them.
2. Given rates: $\frac{dx}{dt} = 35$ mi/h, $\frac{dy}{dt} = 25$ mi/h. Angle $\theta = 60°$ (constant).
3. Relationship (Law of Cosines):
$$z^2 = x^2 + y^2 - 2xy\cos(\theta).$$
Since $\cos(60°) = 1/2$:
$$z^2 = x^2 + y^2 - xy.$$
4. Calculate distances at $t = 4$ hours:
$$x = 35(4) = 140 \text{ mi}, \qquad y = 25(4) = 100 \text{ mi}.$$

5. Calculate z at $t = 4$:

$$z^2 = 140^2 + 100^2 - (140)(100) = 19600 + 10000 - 14000 = 15600,$$

$$z = \sqrt{15600} \approx 124.9 \text{ mi}.$$

6. Differentiate implicitly:

$$2z\frac{dz}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt} - \left(\frac{dx}{dt}y + x\frac{dy}{dt}\right)$$

7. Solve for $\frac{dz}{dt}$:

$$\frac{dz}{dt} = \frac{1}{2z}\left[2x\frac{dx}{dt} + 2y\frac{dy}{dt} - y\frac{dx}{dt} - x\frac{dy}{dt}\right]$$

8. Substitute values:

$$\begin{aligned} 2z\frac{dz}{dt} &= 2(140)(35) + 2(100)(25) - (100)(35) - (140)(25) \\ &= 9800 + 5000 - 3500 - 3500 = 7800 \end{aligned}$$

$$\frac{dz}{dt} = \frac{7800}{2\sqrt{15600}} \approx \mathbf{31.25} \text{ mi/h}$$

6. Optimization Problems

Q53. Minimum Product Find two numbers whose difference is 100 and whose product is a minimum.

S53

1. Let the numbers be x and $x - 100$.
2. Product: $P(x) = x(x - 100) = x^2 - 100x$.
3. Derivative: $P'(x) = 2x - 100$.
4. Critical number: $P'(x) = 0 \Rightarrow x = 50$.
5. $P''(x) = 2 > 0$, so minimum at $x = 50$.
6. Numbers: **50** and $-$**50**.

Q54. Minimum Sum Two positive numbers have product 100. Find the pair with minimum sum.

S54

1. $xy = 100 \Rightarrow y = 100/x$.
2. $S(x) = x + 100/x$.
3. $S'(x) = 1 - 100/x^2$.
4. $S'(x) = 0 \Rightarrow x = 10$.
5. $S''(x) = 200/x^3 > 0$.
6. Numbers: **10** and **10**.

Q55. Minimum of $x + 1/x$ for $x > 0$.

S55

1. $f(x) = x + 1/x$.
2. $f'(x) = 1 - 1/x^2$.
3. $f'(x) = 0 \Rightarrow x = 1$.
4. Minimum at $x = 1$.

Q56. Maximum Area with Fixed Perimeter A rectangle has perimeter 100. Find the dimensions that maximize area.

S56

1. $2x + 2y = 100 \Rightarrow y = 50 - x$.
2. $A(x) = x(50 - x)$.
3. $A'(x) = 50 - 2x$.
4. $A'(x) = 0 \Rightarrow x = 25$.
5. Maximum at $x = y = 25$.

Q57. Minimum Perimeter with Fixed Area A rectangle has area 1000. Find the dimensions that minimize perimeter.

S57

1. $xy = 1000 \Rightarrow y = 1000/x$.
2. $P(x) = 2x + 2000/x$.
3. $P'(x) = 2 - 2000/x^2$.
4. $P'(x) = 0 \Rightarrow x = 10\sqrt{10}$.
5. Minimum when $x = y = 10\sqrt{10}$.

Q58. Closest Point on a Line Find the point on the line $y = 2x + 3$ closest to $(1, 0)$.

S58

1. $D(x) = (x - 1)^2 + (2x + 3)^2$.
2. $D'(x) = 10x + 10$.
3. $D'(x) = 0 \Rightarrow x = -1$.
4. Closest point: $(\mathbf{-1}, \mathbf{1})$.

Q59. Inscribed Cylinder in a Sphere Find the cylinder of maximum volume that can be inscribed in a sphere of radius r.

S59

1. Consider a sphere of radius r centered at the origin. Let the cylinder be symmetric about the x–axis. Let x be **half the height** of the cylinder and y be the radius of its circular base. Then any point on the circle formed by the cross–section satisfies

$$x^2 + y^2 = r^2.$$

2. The full height of the cylinder is $2x$ and the base radius is y, so the volume is
$$V = \pi y^2(2x) = 2\pi x y^2.$$
Using the relation $y^2 = r^2 - x^2$ from the sphere,
$$V(x) = 2\pi x(r^2 - x^2).$$

3. Differentiate $V(x)$ with respect to x:
$$V'(x) = 2\pi\big(r^2 - x^2\big) + 2\pi x(-2x) = 2\pi(r^2 - x^2 - 2x^2) = 2\pi(r^2 - 3x^2).$$

4. Find critical points by setting $V'(x) = 0$:
$$2\pi(r^2 - 3x^2) = 0 \quad \Rightarrow \quad r^2 - 3x^2 = 0 \quad \Rightarrow \quad x = \frac{r}{\sqrt{3}}$$
(we take $x > 0$).

5. Check that this is a maximum: for large x close to r, the volume tends to 0, so the critical point corresponds to a maximum.

6. Substitute $x = r/\sqrt{3}$ back into $V(x)$:
$$V_{\max} = 2\pi \frac{r}{\sqrt{3}}\left(r^2 - \frac{r^2}{3}\right) = 2\pi \frac{r}{\sqrt{3}} \cdot \frac{2r^2}{3} = \boldsymbol{\frac{4\pi r^3}{3\sqrt{3}}}.$$

Q60. Largest Rectangle in an Ellipse Find the rectangle of maximum area that can be inscribed in the ellipse
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

S60

1. By symmetry, it is enough to consider a rectangle whose vertices are (x, y), $(-x, y)$, $(-x, -y)$, $(x, -y)$ in the ellipse with $x > 0$, $y > 0$. Then the area is
$$A = (2x)(2y) = 4xy.$$

2. From the ellipse equation we can solve y in terms of x:
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \Rightarrow \quad \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} \quad \Rightarrow \quad y = b\sqrt{1 - \frac{x^2}{a^2}}.$$
(We take the positive root since $y > 0$.)

3. Substitute into A:
$$A(x) = 4x \cdot b\sqrt{1 - \frac{x^2}{a^2}} = 4b\,x\sqrt{1 - \frac{x^2}{a^2}}.$$
We will maximize $A(x)$ for $0 < x < a$.

4. Differentiate $A(x)$ using the product and chain rules. It is convenient to write
$$A(x) = 4b\,x\left(1 - \frac{x^2}{a^2}\right)^{1/2}.$$
Then
$$A'(x) = 4b\left(1 - \frac{x^2}{a^2}\right)^{1/2} + 4b\,x \cdot \frac{1}{2}\left(1 - \frac{x^2}{a^2}\right)^{-1/2}\left(-\frac{2x}{a^2}\right).$$
Simplifying,
$$A'(x) = 4b\left(1 - \frac{x^2}{a^2}\right)^{1/2} - \frac{4bx^2}{a^2}\left(1 - \frac{x^2}{a^2}\right)^{-1/2}.$$
Factor out $4b\left(1 - \frac{x^2}{a^2}\right)^{-1/2}$:
$$A'(x) = 4b\left(1 - \frac{x^2}{a^2}\right)^{-1/2}\left(1 - \frac{x^2}{a^2} - \frac{x^2}{a^2}\right) = 4b\left(1 - \frac{x^2}{a^2}\right)^{-1/2}\left(1 - \frac{2x^2}{a^2}\right).$$

5. Set $A'(x) = 0$ (and ignore the nonzero factor):

$$1 - \frac{2x^2}{a^2} = 0 \quad \Rightarrow \quad x^2 = \frac{a^2}{2} \quad \Rightarrow \quad x = \frac{a}{\sqrt{2}} \quad (x > 0).$$

   Then from the ellipse equation,

$$y = b\sqrt{1 - \frac{x^2}{a^2}} = b\sqrt{1 - \frac{1}{2}} = \frac{b}{\sqrt{2}}.$$

6. Dolayısıyla, maksimum alanı veren dikdörtgenin boyutları

$$2x = \sqrt{2}\,a, \qquad 2y = \sqrt{2}\,b$$

   ve alanı

$$A_{\max} = 4xy = 4 \cdot \frac{a}{\sqrt{2}} \cdot \frac{b}{\sqrt{2}} = \mathbf{2ab}.$$

Q61. Find the point(s) on the parabola $\mathbf{y^2} = \mathbf{2x}$ closest to the point $(1, 4)$.

S61

Minimize $D^2(y) = (y^2/2 - 1)^2 + (y - 4)^2$. Then

$$\frac{d}{dy}(D^2) = y^3 - 8.$$

The critical point is $y = 2$; then $x = y^2/2 = 2$. Thus the closest point is $(\mathbf{2}, \mathbf{2})$.

Q62. Find the production level x that minimizes $\overline{C}(x) = 100/x + 5 + 0.1x$.

S62

$$\overline{C}'(x) = -\frac{100}{x^2} + 0.1.$$

Setting $\overline{C}'(x) = 0$ gives $x^2 = 1000$, so $x = \mathbf{10\sqrt{10}}$.

ADVANCED CURVE SKETCHING ANALYSIS

Q63. Use curve sketching guidelines to analyze $y = 3\sin x - \sin^3 x$.

S63

(A) **Domain and Periodicity:** The domain is $D = \mathbb{R}$. The function is periodic with period 2π.

(B) **Asymptotes/Intercepts:** No asymptotes (periodic). Intercepts occur when $3\sin x - \sin^3 x = \sin x(3 - \sin^2 x) = 0$. Since $3 - \sin^2 x \neq 0$, intercepts occur where $\sin x = 0$, i.e., at $x = n\pi$.

(C) **Derivatives:**

$$y' = 3\cos x - 3\sin^2 x\cos x = 3\cos x(1 - \sin^2 x) = \mathbf{3\cos^3 x}$$

$$y'' = \frac{d}{dx}(3\cos^3 x) = -\mathbf{9\cos^2 x\sin x}$$

(D) **Critical Points and Monotonicity:** $y' = 0$ when $\cos x = 0$, so $x = \pi/2 + n\pi$.

(E) **Concavity and Inflection Points (IP):** $y'' = 0$ when $\sin x = 0$ or $\cos x = 0$. Concavity changes (IPs) at $x = n\pi$ (where $\sin x$ changes sign); at $x = \pi/2 + n\pi$, y'' does not change sign.

(F) **Local Extrema:** Using the sign of y' around the critical points: in each period, $x = \pi/2 + 2k\pi$ is a local maximum and $x = 3\pi/2 + 2k\pi$ is a local minimum.

(G) **Sign/Shape Table:** On one basic interval $(0, \pi)$ the behavior is:

x	$(0, \frac{\pi}{2})$	$\frac{\pi}{2}$	$(\frac{\pi}{2}, \pi)$
y'	$+$	0	$-$
y''	$-$	$-$	$-$
y-behavior	$\nearrow$	max	$\searrow$
Concavity	$\frown$	$\frown$	$\frown$

The same pattern repeats with period 2π, with additional inflection points at $x = n\pi$.

(H) **Sketch Placeholder (H):**

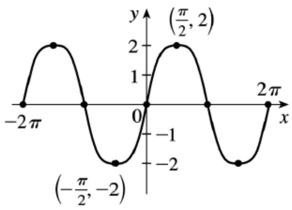

Figure 6: $y = 3\sin x - \sin^3 x$

Q64. Use curve sketching guidelines to analyze $y = e^{2x} - e^x$.

S64

(A) **Domain:** $D = \mathbb{R}$.

(B) **Asymptotes:**

- As $x \to \infty$, $y \to \infty$.
- As $x \to -\infty$, $y \to 0$. Horizontal Asymptote (HA): $\mathbf{y = 0}$.

(C) **Intercepts:** $e^{2x} - e^x = e^x(e^x - 1) = 0$. Since $e^x > 0$, $e^x = 1$, so $\mathbf{x = 0}$ is the intercept. $y(0) = 0$.

(D) **Derivatives:**

$$y' = 2e^{2x} - e^x = \mathbf{e^x(2e^x - 1)}$$
$$y'' = 4e^{2x} - e^x = \mathbf{e^x(4e^x - 1)}$$

(E) **Critical Points and Monotonicity:** $y' = 0$ when $2e^x = 1$, so $x = \ln(1/2) = -\ln \mathbf{2}$. Decreasing on $(-\infty, -\ln 2)$ and increasing on $(-\ln 2, \infty)$.

(F) **Concavity and IPs:** $y'' = 0$ when $4e^x = 1$, so $x = \ln(1/4) = \mathbf{-2}\ln\mathbf{2}$. Concave down on $(-\infty, -2\ln 2)$ and concave up on $(-2\ln 2, \infty)$. IP at $x = -2\ln 2$.

(G) **Local Extrema:** y has a local minimum at $x = -\ln 2$. The value is $y(-\ln 2) = (1/2)^2 - (1/2) = 1/4 - 1/2 = \mathbf{-1/4}$.

(H) **Sign/Shape Table:**

x	$(-\infty, -2\ln 2)$	$-2\ln 2$	$(-2\ln 2, -\ln 2)$	$-\ln 2$	$(-\ln 2, \infty)$
y'	$-$		$-$	0	$+$
y''	$-$	0	$+$	$+$	$+$
y-behavior	$\searrow$	IP	$\searrow$	min	$\nearrow$
Concavity	$\frown$		$\smile$	$\smile$	$\smile$

(I) **Sketch Placeholder (H):**

Q65. Use curve sketching guidelines to analyze $y = xe^{-x}$.

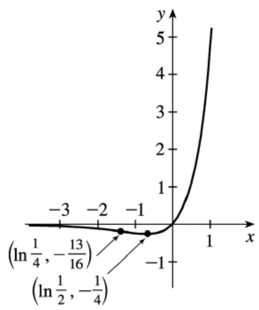

Figure 7: $y = e^{2x} - e^{x}$

S65

(A) Domain: $\mathbb{R}$.

(B) Derivatives:

$$y' = e^{-x}(1 - x), \qquad y'' = e^{-x}(x - 2)$$

(C) CP: $x = 1$ (local max).

(D) IP: $x = 2$.

(E) Table:

x	$(-\infty, 1)$	1	$(1, 2)$	2	$(2, \infty)$
y'	$+$	0	$-$	$-$	$-$
y''	$-$	$-$	$-$	0	$+$
y	$\nearrow$	max	$\searrow$	$\searrow$	$\searrow$
Concavity	$\frown$	$\frown$	$\frown$	IP	$\smile$

(F) Sketch:

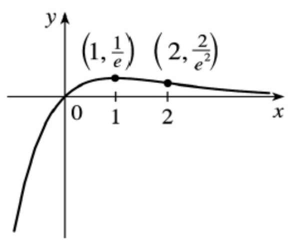

Figure 8: $y = xe^{-x}$

Q66. Use curve sketching guidelines to analyze $y = \ln(\sin x)$.

S66

(A) Domain: $(2n\pi, (2n + 1)\pi)$.

(B) Derivatives:

$$y' = \cot x, \qquad y'' = -\csc^2 x < 0$$

(C) Always concave down.

(D) Table on $(0, \pi)$:

x	$(0, \pi/2)$	$\pi/2$	$(\pi/2, \pi)$
y'	$+$	0	$-$
y''	$-$	$-$	$-$
y	$\nearrow$	max	$\searrow$
Concavity	$\frown$	$\frown$	$\frown$

(E) Sketch:

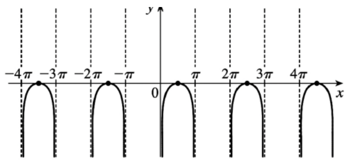

Figure 9: $y = \ln(\sin x)$

Q67. Does $\mathbf{y = x^2/(x^2 - 1)}$ have any oblique asymptotes?

S67

oblique asymptotes occur when the degree of the numerator is one more than the degree of the denominator. Here, both degrees are 2, so there is a horizontal asymptote at $y = 1$ but no oblique asymptotes.

Q68. Sketch the basic shape of $y = x^4 - 4x^3$ near its local minimum.

S68

The local minimum occurs at $x = 3$. Since $y''(3) = 36 > 0$, the graph is concave upward (opening up) near this point.